

● EASY

● ALGEBRA

Systems of two linear equations in two variables

04

6 questions · ~9 min

Q1

ALGEBRA

Connor has c dollars and Maria has m dollars. Connor has 4 times as many dollars as Maria, and together they have a total of \$ 25.00. Which system of equations represents this situation?

- A. $c = 4m$
 $c + m = 25$
- B. $m = 4c$
 $c + m = 25$
- C. $c = 25m$
 $c + m = 4$
- D. $m = 25c$
 $c + m = 4$

A dance teacher ordered outfits for students for a dance recital. Outfits for boys cost \$26, and outfits for girls cost \$35. The dance teacher ordered a total of 28 outfits and spent \$881. If b represents the number of outfits the dance teacher ordered for boys and g represents the number of outfits the dance teacher ordered for girls, which of the following systems of equations can be solved to find b and g ?

A. $26b + 35g = 28$
 $b + g = 881$

B. $26b + 35g = 881$
 $b + g = 28$

C. $26g + 35b = 28$
 $b + g = 881$

D. $26g + 35b = 881$
 $b + g = 28$

$$y = 4$$

$$x = y + 6$$

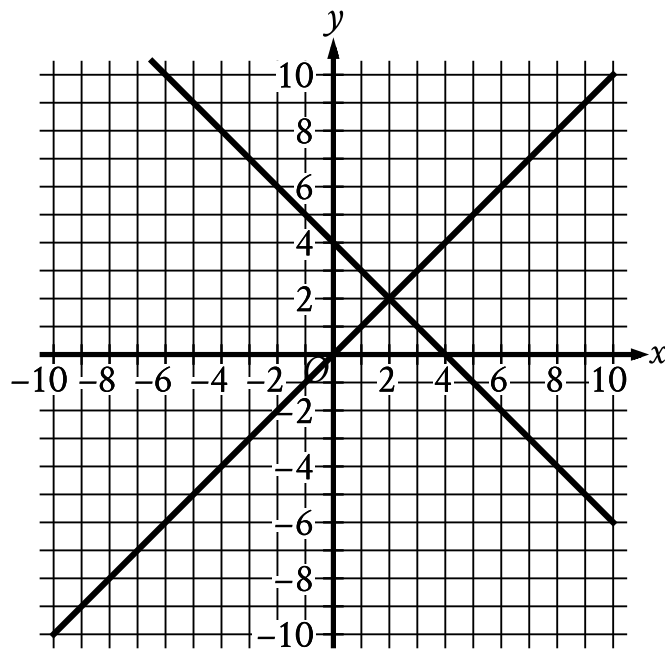
The solution to the given system of equations is x, y . What is the value of x ?

A. 10

B. 6

C. 4

D. 2



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 4)
 - (2 comma 2)
 - (4 comma 0)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma 0)
 - (2 comma 2)
 - (4 comma 4)

The graph of a system of two linear equations is shown. What is the solution x, y to the system?

A. 0, 4

B. 2,2

C. 4,0

D. 4,4

$$x = 5$$

$$y = x - 8$$

Which of the following points x, y is the solution to the given system of equations in the xy -plane?

- A. 0,0
- B. 5,-3
- C. 5,-8
- D. 5,8

$$3x = 12$$

$$-3x + y = -6$$

The solution to the given system of equations is x, y . What is the value of y ?

- A. -3
- B. 6
- C. 18
- D. 30